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Percentage Part 2

Assignment with Solutions

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1. In a school, there are 600 students out of whom 20 percent are girls. Each boy's monthly fee is Rs. 250 and each girl's monthly fee is 30 per cent less than that of a boy. What is the total monthly fee of all the girls and boys together?

- 1) Rs. 1,35,000
- 2) Rs.1,45,000
- 3) Rs.1,41,000
- 4) Rs.1,30,000
- 5) None of these

Answer: 3) Rs.1,41,000

2. The population of a city is 75,000. It increases by 20% during the first year and increased by 30% during second year and decreased by 20% during the third year. What is the population after the third year?

- 1) 1,00,000
- 2) 93,600
- 3) 25,000
- 4) 3,00,000
- 5) None of these

Answer: - 2) 93,600

3. The population of a village increase at the rate of 10% during the first year and decreases at the rate of 20% during the second year. If it has 17,600 inhabitants at present then find the number of inhabitants two years ago?

- 1) 35,000
- 2) 20,000
- 3) 25,000
- 4) 30,000
- 5) None of these

Ans: 2) 20,000

4. The price of sugar is decrease by 10% due to this a family purchase 8kg more sugar for Rs. 720, then find the original consumption?

- 1) 70
- 2) 72
- 3) 75
- 4) 48
- 5) 50

Ans: 2) 72

5. The price of sugar is decrease by 10% due to this a family purchase 8kg more sugar for Rs. 720, then find the current consumption?

- 1) 70
- 2) 72
- 3) 75
- 4) 80
- 5) 50

Ans: 4) 80

6. The price of Rice is increase by 20% due to this a family purchase 5kg less sugar for Rs.250, then find the current price?

- 1) 8 Rs/kg
- 2) 10 Rs/kg
- 3) 15 Rs/kg
- 4) 12 Rs/kg
- 5) 14 Rs/kg

Ans: 2) 10 Rs/kg

7. The price of Rice is increase by 20% due to this a family purchase 5kg less sugar for Rs.240, then find the original price?

- 1) 8 Rs/kg
- 2) 10 Rs/kg
- 3) 15 Rs/kg
- 4) 12 Rs/kg
- 5) 14 Rs/kg

Ans: 2) 10 Rs/kg

8. If the numerator of a fraction is increased by 100% and the denominator is increased by 150%, the resultant fraction is $\frac{3}{5}$. What was the original fraction?

- 1) $\frac{2}{5}$
- 2) $\frac{5}{4}$
- 3) $\frac{7}{4}$
- 4) $\frac{3}{7}$
- 5) $\frac{3}{4}$

Ans:- 5) $\frac{3}{4}$

9. If the numerator of a fraction is increased by 120% and the denominator is increased by 250%, the resultant fraction is $\frac{3}{5}$. What was the original fraction?

- 1) $\frac{3}{4}$
- 2) $\frac{5}{2}$
- 3) $\frac{7}{4}$
- 4) $\frac{3}{8}$
- 5) $\frac{3}{4}$

Ans:- 2) $\frac{5}{2}$

10. In an examination there were $57\frac{1}{7}\%$ girls. Only $12\frac{1}{2}\%$ of the total girls and 25% boys were passed. If number of failed girls be 280 then find the number of failed boys.

1) 150

2) 160

3) 180

4) 220

5) 170

Ans: 3) 180

11. In an examination there were $16\frac{2}{3}\%$ girls. Only $33\frac{1}{3}\%$ of the total girls and 20% boys were passed. If number of failed boys be 120 then find the number of total passed student.

1) 50

2) 60

3) 40

4) 20

5) 70

Ans: 3) 40

12. If the numerator of a fraction is decreased by 20% and the denominator is increased by 50%, the resultant fraction is $\frac{3}{5}$. What was the original fraction?

1) $\frac{3}{7}$

2) $\frac{5}{2}$

3) $\frac{5}{4}$

4) $\frac{2}{3}$

5) $\frac{3}{4}$

Ans: 4) 2/3

13. A reduction of 50 paisa/dozen in the price of eggs a person buys $1\frac{1}{2}$ dozen more eggs for Rs 60. Find original price.

1) 7

2) 6

3) 8

4) 9

5) 10

Ans: 3) 8 Rs.

14. A increment of 1 Rs/dozen in the price of bananas a person buys 2 dozen more bananas for Rs 84. Find new price?

1) 7

2) 6

3) 8

4) 9

5) 10

Ans: 1) 7 Rs.

15. The population of a town is 15400. $\frac{6}{11}$ of them are males and rest females. 33.33% males are married, find the % of female married?

1) 45%

2) 40%

3) 32%

4) 25%

5) 36%

Ans: 2) 40%

Solution

Solution:1-

Total no. of Boys = $600 \times \frac{80}{100} = 480$

$\Rightarrow$ Total monthly Fees of boys = $480 \times 250 = 1,20,000$

$\Rightarrow$ Total no. of Girls = $600 \times \frac{20}{100} = 120$

$\Rightarrow$ Each Girl's monthly fees = $250 \times \frac{70}{100} = 175$

$\Rightarrow$ Total monthly fee's girls = $120 \times 175 = 21,000$

$\Rightarrow$ Total monthly fees of all girls and boys = $1,20,000 + 21,000 = \text{Rs. } 1,41,000$

Solution:2-

Population after 3rd yr. = $75,000 \times 120\% \times 130\% \times 80\% = 93,600$

Solution:3-

Let, the no. of inhabitants 2 yrs. ago = x

$\therefore x \times 110\% \times 80\% = 17,600$

$\therefore x = \frac{(17600 \times 100 \times 100)}{(110 \times 80)} = 20,000$

Solution: 4-

As we discussed in the video ratio of price and quantity will be

In reverse ratio, if expenditure is same.

Here we have 10% decrease in price so it means $10\% = \frac{1}{10} \Rightarrow \frac{9}{10}$

Price = $\frac{10}{9}$

Quantity = $\frac{9}{10}$ so $1 = 8$

Then original consumption will be $9 = 72 \text{ kg}$ Ans.

Solution:5 –

As we discussed in the video ratio of price and quantity will be

In reverse ratio, if expenditure is same.

Here we have 10% decrease in price so it means $10\% = \frac{1}{10} \Rightarrow \frac{9}{10}$

Price = 10 9

Quantity = 9 10 so 1 = 8

Then original consumption will be $10 = 80$ kg Ans.

Solution:6-

As we discussed in the video ratio of price and quantity will be

In reverse ratio, if expenditure is same.

Here we have 20% increase in price so it means $20\% = \frac{1}{5} \Rightarrow \frac{6}{5}$

Price = 5 6

Quantity = 6 5 so 1 = 5

Then current consumption will be $5 = 25$ kg

Then the current price will be $25 = 250 \Rightarrow 10$ Rs/kg.

Solution:7-

As we discussed in the video ratio of price and quantity will be

In reverse ratio, if expenditure is same.

Here we have 20% increase in price so it means $20\% = \frac{1}{5} \Rightarrow \frac{6}{5}$

Price = 5 6

Quantity = 6 5 so 1 = 5

Then original consumption will be $6 = 30$ kg

Then the original price will be $30 = 240 \Rightarrow 8$ Rs/kg.

Solution:8 - $(x \times 200/100)/(y \times 250/100) = 3/5 \Rightarrow$ on solving we get $x/y = 3/4$ Ans.

Solution:9 - $(x \times 220/100)/(y \times 350/100) = 11/7 \Rightarrow$ on solving we get $x/y = 5/2$ Ans

Solution:10 - $57 \frac{1}{7}\% = 4/7$, $12 \frac{1}{2}\% = 1/8$ and $25\% = 1/4$

Let total number of students is 56 which is multiple of 4 and 7

Girls are $4/7 \Rightarrow 32$ then the rest are boys $\Rightarrow 24$

Passed girls are = 4 girls are passed then 28 will be fail

Passed boys are = 6 boys are passed then 18 will be fail

He we have given 28 = 280 means 10 times

Number of failed boys which is 18 will be 180 Ans.

Solution:11 - $16 \frac{2}{3}\% = 1/6$, $33 \frac{1}{3}\% = 1/3$ and $20\% = 1/5$

Let total number of students is 90 which is multiple of 5,6 and 9

Girls are $1/6 \Rightarrow 15$ then the rest are boys $\Rightarrow 75$

Passed girls are = 5 girls are passed then 10 will be fail

Passed boys are = 15 boys are passed then 60 will be fail

He we have given 60 = 120 means 2 times

Number of failed boys which is 20 will be 40 Ans.

Solution:12 - $(x \times 80/100)/(y \times 150/100) = 16/45 \Rightarrow$ on solving we get $x/(y) = 2/(3)$ Ans

Solution:-13- initial price is x then after reduction it will be $\Rightarrow (x - .50)$

$60/(x-.5) - 60/(x) = 1/2 \Rightarrow$ on solving we get $\Rightarrow x = 8$ rupees. Initial price.

Solution:14 - initial price is x then after reduction it will be $\Rightarrow (x - .50)$

$84/x - 84/(x+1) = 2 \Rightarrow$ on solving we get $\Rightarrow x = 6$ rupees. Initial price.

New price will be $6+1 = 7$ Rs. ans.

Solution15: - Total population is 11 there are 6 males and 5 will be females

33.33% of the male are married means 2 are married

If 2 males are married then also 2 females will be married

% Married females will be $\Rightarrow 2/5 \Rightarrow 40\%$